Hackathon Challenge Pack

Common challenge mechanics

- Level 1 (0 - 3 hours): demonstrate a working end-to-end core journey.
- Judging: award points for observable behaviour, not architecture slides or unsupported AI claims. Use synthetic or properly anonymised data only. No production credentials, real customer PII, external payments or live collection calls.
- Organiser responsibility: provide schemas, sample data, hidden tests, credentials for sandbox APIs, evaluation rubric and expected policy/regulatory rules. A prototype is not certified for regulatory compliance.
- No AI resources provided: the organiser will not provide AI model access, API keys, usage credits, model hosting or speech services. Teams must use their own accounts and bear any associated costs.

1. Digital Fraud Investigation System

Category: AI for Fraud Detection, Investigation & Financial Crime Prevention
Applicable Domains: Digital Lending, Insurance, Core Banking and Digital Financial Services
Focus: Fraud Detection, Risk Intelligence, Investigation, Prevention and Case Management

Problem Statement

Financial institutions increasingly rely on digital applications to onboard customers, process loan applications, issue insurance policies, execute financial transactions and deliver banking services. While these digital journeys improve customer experience and operational efficiency, they also introduce opportunities for fraud.

Fraud is no longer limited to suspicious financial transactions. It can occur at different stages of a customer's digital journey, from identity verification and account creation to loan disbursement, insurance claims and fund transfers. Fraudsters may use stolen identities, manipulated documents, compromised accounts, multiple fake profiles, shared devices, coordinated transactions or fabricated insurance claims to exploit financial systems. For financial institutions operating across lending, insurance and core banking, detecting these activities independently within each system is often insufficient. A transaction may appear legitimate in isolation but become suspicious when connected to a customer's previous applications, login behaviour, device history or activities across other financial products.

For example, a customer may apply for multiple loans using different identities but the same device, receive disbursements into related bank accounts and subsequently initiate unusual transfers. Each activity may appear acceptable individually, but the combined behaviour could indicate coordinated fraud.

The challenge is to build an AI-powered Digital Fraud Investigation System that identifies suspicious activities across digital financial journeys, connects related events, assesses fraud risk and enables financial institutions to investigate and respond appropriately. The application must go beyond basic fraud detection based on transaction amounts or predefined thresholds. It should identify suspicious patterns across customer profiles, transactions, digital applications, devices, login activities, lending operations, insurance activities and core banking systems.

The solution must distinguish potentially fraudulent behaviour from legitimate customer activity, explain why an event was flagged, retain supporting evidence and enable authorised users to investigate, escalate or dismiss cases.

Participants are encouraged to introduce additional fraud detection techniques and innovative approaches beyond the specified scenarios. Solutions demonstrating meaningful detection of emerging or sophisticated digital fraud patterns will receive additional consideration within the evaluation framework.

The organiser will assess actual application behaviour rather than a static dashboard, predefined presentation or standalone AI-generated response.

Digital Fraud Scenarios to Be Addressed

The solution should consider the following types of digital fraud. These scenarios represent practical challenges faced by financial institutions and provide the basis for progressive evaluation.

1. Identity Fraud and Synthetic Customer Profiles

Fraudsters may create customer profiles using stolen identification details, altered documents or combinations of genuine and fabricated information. The application should identify inconsistent identity attributes, repeated contact information, suspiciously similar profiles and unusual relationships between supposedly independent customers.

2. Device Sharing and Coordinated Fraud

Multiple customers may use the same device, IP address, contact number or digital identity attributes to submit applications or initiate transactions. The system should identify potentially coordinated activity while recognising legitimate shared devices, such as family members using a common phone.

3. Account Takeover and Suspicious Login Activity

Fraudsters may gain unauthorised access to customer accounts through stolen credentials, phishing or compromised authentication mechanisms. Suspicious behaviour may include repeated failed login attempts, unexpected device changes, unusual access locations, password resets followed by large transfers or changes to beneficiary information.

4. Cross-System and Organised Fraud

Fraud patterns may become visible only when information from lending, insurance and core banking systems is connected. The solution should correlate activity across these systems and identify related customers, accounts, devices, applications, transactions and claims that could indicate organised fraud.

Level 1: Detect, Analyse and Flag Suspicious Digital Activity
Suggested time: 0 -3 hours
Objective: Build a working fraud detection application that ingests digital financial activity, identifies suspicious patterns and presents explainable fraud alerts for investigation.

Required Capabilities
1. Data Ingestion and Integration
Build an application that allows users to upload or select data from different digital financial systems. The application should support the following sample data:

- Customer and account profiles containing customer identifiers, account details, contact information and onboarding information.
- Transaction batches containing amounts, timestamps, sender and beneficiary details, transaction channels and statuses.
- Device and login logs containing device identifiers, IP addresses, login timestamps, authentication outcomes and device changes.

The application should accept common structured formats such as CSV, Excel and JSON. It must validate uploaded records, identify missing mandatory information and avoid silently ignoring invalid data. Participants should use a common data structure that allows records from different systems to be linked through customer, account, application, device or transaction identifiers.
2. Fraud Detection Rules and Behavioural Analysis
Implement a detection mechanism that identifies suspicious activities using configurable rules, statistical analysis, machine learning or a combination of approaches. At a minimum, the application must detect the following four scenarios:

Scenario	Expected detection behaviour
Suspicious transfers	Identify unusual amounts, rapid transfers, new beneficiaries or transactions inconsistent with customer history
Shared devices	Identify multiple customer profiles using the same device and assess whether the relationship is suspicious
Repeated login failures	Detect repeated unsuccessful logins, particularly when followed by a successful login or sensitive account activity
Legitimate lookalikes	Avoid incorrectly classifying ordinary customer behaviour as fraud merely because it resembles a suspicious pattern

Participants should also implement at least one additional fraud scenario from the challenge catalogue, such as suspicious loan applications or duplicate insurance claims.

3. Fraud Risk Assessment
Every detected suspicious activity should receive a risk classification.
The application may use a risk score, such as 0–100, or clearly defined categories such as Low, Medium and High. The score should consider multiple relevant indicators rather than relying only on a single threshold.
For example, a high-value transfer should not automatically be considered fraudulent if the customer regularly performs similar transactions.
However, a high-value transfer immediately following multiple failed login attempts and a new-device login may warrant greater scrutiny. The application must explain the factors contributing to the assigned risk level.
4. Fraud Alert Dashboard
Develop a dashboard where users can inspect detected suspicious activities.
The dashboard should display the number of processed records, flagged events, risk distribution and fraud categories. Users must be able to filter alerts by customer, transaction, product, risk level, date and detection type.

Selecting an alert should display the underlying event, the reasons it was flagged and the supporting records.

5. Basic Fraud Case Queue

Each actionable fraud alert should be available in a case queue.

A case should contain a unique identifier, affected customer or account, suspected fraud category, risk level, supporting evidence and investigation status. Users should be able to open the case and record an initial decision, such as Requires Investigation or False Positive. The system must preserve the original alert even when its classification is subsequently changed.

Expected Submission

A working application demonstrating data ingestion, fraud detection, risk assessment, an alert dashboard and a basic investigation queue.